

EXAMPLE 1 Bacterial Growth

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{1}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

Growth and Decay

1. The population of a community is known to increase at a rate proportional to the number of people present at time t . If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?
2. Suppose it is known that the population of the community in Problem 1 is 10,000 after 3 years. What was the initial population P_0 ? What will be the population in 10 years? How fast is the population growing at $t = 10$?

① Solⁿ: let, $P(t)$ = population at present time t in years.
 $\frac{dP}{dt}$ = Rate of increases.

$$\text{Now, } \frac{dP}{dt} \propto P(t)$$

$$\Rightarrow \frac{dP}{dt} = kP(t)$$

$$\Rightarrow \int \frac{dP}{P} = \int k dt$$

$$\Rightarrow \ln P = kt + c,$$

$$\Rightarrow P = e^{kt + c},$$

$$\therefore P(t) = ce^{kt} \quad \text{--- (i)}$$

$$\text{At } t=0, P=P_0$$

$$\therefore P_0 = ce^0 = c$$

$$\text{Now, } P(t) = P_0 e^{kt} \quad \text{--- (ii)}$$

Given conditions,

$$t = 5, \quad P = 2P_0$$

$$2P_0 = P_0 e^{k \cdot 5}$$

$$\Rightarrow \ln 2 = \ln e^{5k}$$

$$\Rightarrow k = \frac{1}{5} \ln 2$$

$$\text{Now, to triple, } 3P_0 = P_0 e^{\frac{1}{5} \ln 2 \cdot t}$$

$$\Rightarrow t = 7.9248 \text{ years.}$$

$$\text{to quadruple, } 4P_0 = P_0 e^{\frac{1}{5} \ln 2 \cdot t}$$

$$\Rightarrow t = 10 \text{ years.}$$

② From the solution of problem 1,

$$P = P_0 e^{kt}$$

Now, from the given condition,

$$10000 = P_0 e^{\frac{1}{5} \ln 2 \times 3} \quad \Bigg| \quad \text{if the rate } k = \frac{1}{5} \ln 2$$

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$$\Rightarrow P_0 = 6598$$

Now, in 10 years.

$$P = 6598 e^{\frac{1}{5} \ln 2 \times 10}$$

$$= 26392$$

In order to find how fast the population is going,

$$\frac{dP}{dt} = kP$$

$$\Rightarrow \frac{dP}{dt} = \frac{1}{5} \ln 2 \times 26392$$

$$= 3659 / \text{year}.$$